

THE WELCH FUNCTION IS NOT THIRD-ORDER SUM-FREE

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ABSTRACT. Let $n = 2m + 1$. Hou and Zhao conjectured that for $n \geq 7$ the Welch function X^{2^m+3} on $\mathbb{F}_{2^n}$ sums to zero over some three-dimensional affine subspace, that is, it is not third-order sum-free. We prove this. The proof counts exactly the elements x for which some subspace spanned by 1, x and a third element gives a zero sum: there are $(2^n + 1 - 3K_n)/4$ of them, where K_n is a binary Kloosterman sum.

1. INTRODUCTION

A function $f: \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ is *kth order sum-free* if $\sum_{a \in A} f(a) \neq 0$ for every k -dimensional affine subspace A of $\mathbb{F}_{2^n}$ over $\mathbb{F}_2$. Carlet introduced this notion [3, 2]; for $k = 2$ it is the APN property. Throughout, $n = 2m + 1$ is odd, $q = 2^n$, $F = \mathbb{F}_q$ and $\text{Tr}: F \rightarrow \mathbb{F}_2$ is the absolute trace. The *Welch function* is

$$W_n(X) = X^{2^m+3}.$$

It is APN [5], and decimating a binary m -sequence by $2^m + 3$ gives a three-valued cross-correlation, as Welch conjectured [1]. For $m \geq 2$ it has algebraic degree 3, so the first open question about its higher-order sum-freeness is the third order. For $m = 2$ the Welch function is $X^7 = X^{2^3-1}$, which is third-order sum-free by a result of Carlet [3, Proposition 1]. Hou and Zhao [6] reduced the question to a family of linearized polynomials, computed it for $7 \leq n \leq 15$, and proposed the following statement [6, Conjecture 4.10].

Conjecture 1. For $n = 2m + 1 \geq 7$, the Welch function on $\mathbb{F}_{2^n}$ is not third-order sum-free.

We prove Conjecture 1. The reduction of Hou and Zhao goes through the polynomial

$$(1.1) \quad g(X, Y) = Y^{2^m}(X^2 + X) + Y^2(X^{2^m} + X) + Y(X^{2^m} + X^2).$$

For $x \in F$ let $Z(x) = \{y \in F : g(x, y) = 0\}$. Since $g(x, Y)$ is an additive polynomial in Y , $Z(x)$ is a vector space over $\mathbb{F}_2$, and it contains 1 and x . By Lemma 2.3 below, W_n fails to be third-order sum-free exactly when $\dim_{\mathbb{F}_2} Z(x) \geq 3$ for some $x \in F \setminus \mathbb{F}_2$. Our main result decides this dimension for every x .

Theorem 1.1. Let $n = 2m + 1 \geq 3$ and $x \in F \setminus \mathbb{F}_2$, and put $u = x^{2^m}$. Then $\dim_{\mathbb{F}_2} Z(x) \in \{2, 3\}$, and $\dim_{\mathbb{F}_2} Z(x) = 3$ if and only if

$$\text{Tr}\left(\frac{1}{x+u}\right) = 1 \quad \text{and} \quad \text{Tr}\left(\frac{xu}{x+u}\right) = 1.$$

Let

$$K_n = \sum_{z \in F^*} (-1)^{\text{Tr}(z+z^{-1})}$$

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be the Kloosterman sum of F . The conditions of Theorem 1.1 turn the count of the elements x with $\dim Z(x) = 3$ into Kloosterman sums.

Theorem 1.2. *Let $n = 2m+1 \geq 3$, and let N_n be the number of $x \in F \setminus \mathbb{F}_2$ with $\dim_{\mathbb{F}_2} Z(x) = 3$. Then*

$$N_n = \frac{2^n + 1 - 3K_n}{4}.$$

By Weil's bound [8] (see also [7, Chapter 5]), $|K_n| \leq 2 \cdot 2^{n/2}$. Hence

$$N_n \geq \frac{2^n + 1 - 6 \cdot 2^{n/2}}{4} = \frac{(2^{n/2} - 3)^2 - 8}{4} > 0 \quad \text{for } n \geq 7.$$

Corollary 1.3. *Conjecture 1 is true. More precisely, for $n = 2m+1 \geq 5$ the Welch function sums to zero over exactly*

$$\frac{N_n(2^n - 1)}{42}$$

three-dimensional linear subspaces of F , and over exactly 2^{n-3} times as many three-dimensional affine subspaces. This number is positive for every $n \geq 7$ and is zero for $n = 5$.

By a theorem of Carlitz [4] (see also [7, Chapter 5]), $K_n = -(\omega^n + \bar{\omega}^n)$ with $\omega = (-1 + \sqrt{-7})/2$, because $K_1 = 1$. So $K_0 = -2$, $K_1 = 1$ and $K_n = -K_{n-1} - 2K_{n-2}$, which gives Table 1. Hou and Zhao [6, Corollary 4.9] showed that $\dim Z(x) = 3$ for some $x \in F \setminus \mathbb{F}_2$ if and only if the polynomial $\det D^*(X)$ built from a Dickson matrix has more than two roots in F ; the number of its roots in F is $2 + N_n$. For $5 \leq n \leq 15$ these are the values Hou and Zhao computed [6, Table 1].

TABLE 1. The Kloosterman sum K_n , the count N_n of Theorem 1.2 and the number $2 + N_n$ of roots of $\det D^*(X)$ in $\mathbb{F}_{2^n}$.

n	K_n	N_n	$2 + N_n$
5	11	0	2
7	-13	42	44
9	-5	132	134
11	67	462	464
13	-181	2184	2186
15	275	7986	7988
17	-101	32844	32846
19	-797	131670	131672
21	2795	522192	522194

The proof of Theorem 1.1 uses the automorphism $\rho(z) = z^{2^{m+1}}$ of F , whose square is the Frobenius map $z \mapsto z^2$ and whose inverse is $z \mapsto z^{2^m}$. Applying ρ to the equation $g(x, y) = 0$ and eliminating y^{2^m} produces an additive polynomial of degree 8 in y . Its roots form a three-dimensional space V that always lies in F and contains $Z(x)$, and a linear map built from $g(x, \cdot)$ moves V into $Z(x)$. Whether $Z(x) = V$ is then decided by two elements of $\mathbb{F}_2$, and both turn out to be traces of explicit rational functions of x and u , because an element of $\mathbb{F}_2$ equals its own trace when n is odd.

Section 2 collects facts about the trace and the reduction of Hou and Zhao. Section 3 studies a fixed x , Section 4 proves Theorem 1.1, and Section 5 proves Theorem 1.2 and Corollary 1.3.

2. PRELIMINARIES

Lemma 2.1. *Let $\rho(z) = z^{2^{m+1}}$ for $z \in F$.*

- (a) $\rho(z^{2^m}) = z = \rho(z)^{2^m}$ and $\rho(\rho(z)) = z^2$ for every $z \in F$.
- (b) *The elements of F fixed by $z \mapsto z^{2^m}$, and those fixed by ρ , are 0 and 1.*

Proof. (a) holds because $z^{2^n} = z$, $2^m \cdot 2^{m+1} = 2^n$ and $2^{m+1} \cdot 2^{m+1} = 2 \cdot 2^n$. For (b), the fixed field of $z \mapsto z^{2^k}$ is $\mathbb{F}_{2^{\gcd(k,n)}}$, and $\gcd(m, 2m+1) = \gcd(m+1, 2m+1) = 1$. $\square$

Let $T_0 = \{w \in F : \text{Tr}(w) = 0\}$, put $\wp(z) = z^2 + z$, and for $z \in F$ let

$$T_m(z) = \sum_{i=0}^{m-1} z^{2^i}.$$

Lemma 2.2. (a) $\text{Tr}(z^{2^k}) = \text{Tr}(z)$ for all $z \in F$ and $k \geq 0$; in particular $\text{Tr}(\rho(z)) = \text{Tr}(z)$.

- (b) $\text{Tr}(1) = 1$, so $\text{Tr}(\varepsilon) = \varepsilon$ for $\varepsilon \in \mathbb{F}_2$.
- (c) *Each of the maps $z \mapsto \wp(z)$, $z \mapsto z^{2^m} + z$ and $z \mapsto \rho(z) + z$ is $\mathbb{F}_2$ -linear with kernel $\mathbb{F}_2$ and image T_0 .*
- (d) $\wp(T_m(z)) = T_m(\wp(z)) = z^{2^m} + z$ for all $z \in F$.

Proof. (a) is clear and (b) holds because n is odd. In (c), the kernels are $\mathbb{F}_2$ by Lemma 2.1(b), so each image has $q/2$ elements; each image lies in T_0 by (a), and $|T_0| = q/2$. For (d), both sides telescope: $\wp(T_m(z)) = \sum_{i=1}^m z^{2^i} + \sum_{i=0}^{m-1} z^{2^i} = z^{2^m} + z$, and $T_m(z^2 + z)$ is the same sum. $\square$

The next lemma is the reduction of Hou and Zhao [6, Section 4]; we include the short proof.

Lemma 2.3. (a) *For $x, y \in F$,*

$$\sum_{a_1, a_2, a_3 \in \mathbb{F}_2} W_n(a_1x + a_2y + a_3) = g(x, y).$$

- (b) *Let $A = c + A_0$ be a three-dimensional affine subspace of F with linear part A_0 . Then $\sum_{a \in A} W_n(a) = \sum_{a \in A_0} W_n(a)$, and $\sum_{a \in A_0} W_n(a) = b^{2^m+3} \sum_{a \in b^{-1}A_0} W_n(a)$ for every $b \in A_0 \setminus \{0\}$.*
- (c) *W_n is not third-order sum-free if and only if $\dim_{\mathbb{F}_2} Z(x) \geq 3$ for some $x \in F \setminus \mathbb{F}_2$.*

Proof. (a) Since $a_i^{2^k} = a_i$ for $a_i \in \mathbb{F}_2$,

$$W_n(a_1x + a_2y + a_3) = (a_1x^{2^m} + a_2y^{2^m} + a_3)(a_1x^2 + a_2y^2 + a_3)(a_1x + a_2y + a_3).$$

Expanding, we get a sum of terms $a_i a_j a_k \alpha \beta \gamma$ with $\alpha \in \{x^{2^m}, y^{2^m}, 1\}$, $\beta \in \{x^2, y^2, 1\}$ and $\gamma \in \{x, y, 1\}$. Summed over $(a_1, a_2, a_3) \in \mathbb{F}_2^3$, a monomial in the a_i gives 0 unless all three of a_1, a_2, a_3 occur in it, because $\sum_{a \in \mathbb{F}_2} 1 = 0$. So only the six terms in which i, j, k are distinct survive, each with sum 1, and they add up to $g(x, y)$.

(b) Write the elements of A as $c + a$ with $a \in A_0$ and expand $W_n(c + a) = (c^{2^m} + a^{2^m})(c^2 + a^2)(c + a)$. Apart from $a^{2^m} a^2 a$, every one of the eight products is a constant times a product of at most two $\mathbb{F}_2$ -linear functions of a . In coordinates $a = a_1 e_1 + a_2 e_2 + a_3 e_3$ with respect to a basis of A_0 , such a product is a combination of monomials in at most two of the a_i , and its sum over A_0 vanishes as in (a). The second statement holds because W_n is a power function.

(c) By (b), W_n is not third-order sum-free if and only if it sums to zero over a three-dimensional linear subspace that contains 1. Such a subspace is $\langle 1, x, y \rangle$ with $1, x, y$ linearly independent over $\mathbb{F}_2$, and by (a) the sum over it is $g(x, y)$. Since $Z(x)$ always contains $\langle 1, x \rangle$, a suitable y exists if and only if $\dim Z(x) \geq 3$. $\square$

3. A FIXED ELEMENT x

In this section and the next, $x \in F \setminus \mathbb{F}_2$ is fixed and

$$u = x^{2^m}, \quad t = x^2 + x, \quad d = x + u, \quad v = u^2 + u.$$

By Lemma 2.1, $\rho(x) = u^2$ and $\rho(u) = x$, so

$$(3.1) \quad \rho(t) = v^2, \quad \rho(d) = d + v = x + u^2, \quad \rho(d + v) = d^2, \quad \rho(x^2 + u) = u^4 + x, \quad t^{2^m} = v.$$

The elements t , $u^2 + u = v$, d and $d + v = x + \rho(x)$ are nonzero by Lemma 2.1(b), and

$$(3.2) \quad t = d^2 + d + v.$$

Write $L(y) = g(x, y)$, so that

$$L(y) = t y^{2^m} + d y^2 + (x^2 + u) y,$$

an $\mathbb{F}_2$ -linear map of F with kernel $Z(x)$. One checks directly that $L(1) = L(x) = 0$.

Lemma 3.1. $L(y) = t T_m(\wp(y)) + d \wp(y)$ for every $y \in F$.

Proof. By Lemma 2.2(d), $t T_m(\wp(y)) + d \wp(y) = t(y^{2^m} + y) + d(y^2 + y) = t y^{2^m} + d y^2 + (t + d) y$, and $t + d = x^2 + u$. $\square$

Let

$$Y(y) = \frac{d y^2 + (x^2 + u) y}{t}, \quad P(y) = v^2 y + (d + v) Y(y)^4 + (u^4 + x) Y(y)^2,$$

and define $\Phi: F \rightarrow F$ by $\Phi(y) = \rho(L(y)/t)$. Then Y , P and Φ are $\mathbb{F}_2$ -linear, $L(y) = t(y^{2^m} + Y(y))$, and $\Phi(y) = 0$ if and only if $L(y) = 0$.

Lemma 3.2. $P(y) = \rho(L(\Phi(y)))$ for every $y \in F$.

Proof. Let $\ell = L(y)/t$, so that $y^{2^m} = Y(y) + \ell$ and, by Lemma 2.1(a), $\rho(y) = (y^{2^m})^2 = (Y(y) + \ell)^2$. Applying ρ to $L(y)$ and using (3.1),

$$\rho(L(y)) = v^2 y + (d + v) \rho(y)^2 + (u^4 + x) \rho(y) = P(y) + (d + v) \ell^4 + (u^4 + x) \ell^2.$$

On the other hand $\rho(L(y)) = \rho(t\ell) = v^2 \rho(\ell)$. Hence $P(y) = v^2 \rho(\ell) + (d + v) \ell^4 + (u^4 + x) \ell^2$. Now put $\ell' = \Phi(y) = \rho(\ell)$. By Lemma 2.1(a), $\ell'^{2^m} = \ell$ and $\rho(\ell') = \ell^2$, so

$$\rho(L(\ell')) = \rho(t\ell + d\ell'^2 + (x^2 + u)\ell') = v^2 \ell' + (d + v) \ell^4 + (u^4 + x) \ell^2 = P(y). \quad \square$$

Let $\Pi(y) = y(y+1)(y+x)(y+x+1) = \wp(y)(\wp(y)+t) = y^4 + (t+1)y^2 + ty$. It is an additive polynomial whose roots are the elements of $\langle 1, x \rangle$. Put

$$\lambda = \frac{(d+v)d^4}{t^4}, \quad c = \frac{t^3 v^2}{d^4(d+v)}, \quad a = \frac{c}{t^2} = \frac{t v^2}{d^4(d+v)}.$$

Lemma 3.3. As polynomials in y over F , $P(y) = \lambda \Pi(y)(\Pi(y) + c)$. The roots of P in an algebraic closure of F form a three-dimensional $\mathbb{F}_2$ -space V that contains $\langle 1, x \rangle$, and $V \setminus \langle 1, x \rangle = \{y : \Pi(y) = c\}$.

Proof. P is an additive polynomial. Its coefficient of y^8 is $\lambda \neq 0$ and its coefficient of y is $v^2 \neq 0$, since $Y(y)^4$ and $Y(y)^2$ contain only y^8 , y^4 and y^2 . So P has 8 distinct roots, and they form a three-dimensional $\mathbb{F}_2$ -space V . Since $Y(1) = 1$ and $Y(x) = u$,

$$P(1) = v^2 + (d + v) + (u^4 + x) = 0, \quad P(x) = v^2 x + (d + v)u^4 + (u^4 + x)u^2 = 0,$$

so $\langle 1, x \rangle \subseteq V$. Choose $y_0 \in V \setminus \langle 1, x \rangle$. Then $V = \langle 1, x \rangle \cup (y_0 + \langle 1, x \rangle)$ and, since Π is additive,

$$P(y) = \lambda \Pi(y) \Pi(y + y_0) = \lambda \Pi(y) (\Pi(y) + \Pi(y_0)).$$

The coefficient of y on the right is $\lambda t \Pi(y_0)$, because $\Pi(y)^2$ has only even powers of y . Comparing with v^2 gives $\Pi(y_0) = v^2/(\lambda t) = c$. Finally $c \neq 0$, so the roots of $\Pi(y) + c$ are exactly the four elements of V outside $\langle 1, x \rangle$. $\square$

Lemma 3.4. $\text{Tr}(a) = 0$. If $z \in F$ satisfies $\wp(z) = a$, then $\text{Tr}(tz) = 0$. Consequently $V \subseteq F$.

Proof. From (3.2) one checks the identity

$$(3.3) \quad a = \wp\left(\frac{v}{d^2}\right) + \frac{1}{d} + \frac{1}{d+v};$$

multiplied by $d^4(d+v)$ it reads $tv^2 = v^2(d+v) + vd^2(d+v) + d^3(d+v) + d^4$, and the right side equals $v^2(d^2 + d + v)$. By (3.1), $1/(d+v) = \rho(1/d)$, so $\text{Tr}(1/(d+v)) = \text{Tr}(1/d)$ and $\text{Tr}(a) = 0$. By Lemma 2.2(c) there are exactly two elements $z \in F$ with $\wp(z) = a$, and they differ by 1; since $\text{Tr}(t) = \text{Tr}(\wp(x)) = 0$, the value $\text{Tr}(tz)$ does not depend on the choice.

Let $h = 1/d$. Since $\text{Tr}(h + \text{Tr}(h)) = 0$ by Lemma 2.2(b), Lemma 2.2(c) gives $z_2 \in F$ with $\rho(z_2) + z_2 = h + \text{Tr}(h)$. By Lemma 2.1(a),

$$\wp(z_2) = \rho(\rho(z_2)) + z_2 = \rho(\rho(z_2) + z_2) + (\rho(z_2) + z_2) = \rho(h) + h = \frac{1}{d+v} + \frac{1}{d},$$

so by (3.3) we may take $z = v/d^2 + z_2$. First, by (3.2),

$$\frac{tv}{d^2} = v + \frac{v}{d} + \frac{v^2}{d^2} = \wp(u) + \wp\left(\frac{v}{d}\right),$$

which has trace 0. Second, let $e = d + v = x + \rho(x)$. Then $\rho(e) = d^2$ by (3.1), and $\rho(e) + e = x^2 + x = t$. Using Lemma 2.2(a) and Lemma 2.1(a) to move $z \mapsto z^{2^m}$ and ρ through the trace,

$$\begin{aligned} \text{Tr}(tz_2) &= \text{Tr}(\rho(e)z_2) + \text{Tr}(ez_2) = \text{Tr}(e z_2^{2^m}) + \text{Tr}(e z_2) = \text{Tr}(e (\rho(z_2) + z_2)^{2^m}) \\ &= \text{Tr}(\rho(e) (\rho(z_2) + z_2)) = \text{Tr}(d^2(h + \text{Tr}(h))) = \text{Tr}(d) + \text{Tr}(h) \text{Tr}(d^2) = 0, \end{aligned}$$

since $\text{Tr}(d) = \text{Tr}(x + x^{2^m}) = 0$. Hence $\text{Tr}(tz) = 0$.

Now Lemma 2.2(c) gives $y \in F$ with $\wp(y) = tz$. Then $\Pi(y) = tz(tz + t) = t^2\wp(z) = c$, so $y \in V \setminus \langle 1, x \rangle$ by Lemma 3.3, and $V = \langle 1, x, y \rangle \subseteq F$. $\square$

Lemma 3.5. $V = \{y \in F : L(\Phi(y)) = 0\}$, $Z(x) \subseteq V$ and $\Phi(V) \subseteq Z(x)$. The element $\delta(x) = \Phi(y_0)$ is the same for all $y_0 \in V \setminus \langle 1, x \rangle$ and lies in $\langle 1, x \rangle$. If $\delta(x) = 0$ then $Z(x) = V$; otherwise $Z(x) = \langle 1, x \rangle$.

Proof. Since $V \subseteq F$ (Lemma 3.4) and ρ is injective, Lemma 3.2 shows that $V = \{y \in F : L(\Phi(y)) = 0\}$. If $L(y) = 0$ then $\Phi(y) = 0$ and $y \in V$, so $Z(x) \subseteq V$; and $\Phi(V) \subseteq Z(x)$ by the same description of V . Two choices of y_0 differ by an element of $\langle 1, x \rangle \subseteq Z(x)$, on which Φ vanishes. As $\langle 1, x \rangle \subseteq Z(x) \subseteq V$ and $\dim V = 3$, either $Z(x) = V$ or $Z(x) = \langle 1, x \rangle$. In the first case $\delta(x) = 0$. In the second case $\delta(x) \in \Phi(V) \subseteq \langle 1, x \rangle$, and $\delta(x) \neq 0$ because $y_0 \notin Z(x)$. $\square$

Lemma 3.6. $\delta(x) \in \{0, 1\}$ if and only if $\text{Tr}(1/d) = 1$.

Proof. Take $y_0 \in V \setminus \langle 1, x \rangle$ and let $z = \wp(y_0)/t$. By Lemma 3.3, $t^2\wp(z) = \wp(y_0)(\wp(y_0) + t) = c$, so $\wp(z) = a$. Let $\ell_0 = L(y_0)/t$, so that $\delta(x) = \rho(\ell_0)$, and $\delta(x) \in \{0, 1\}$ if and only if $\ell_0 \in \{0, 1\}$, that is, $\wp(\ell_0) = 0$. By Lemma 3.1, $\ell_0 = T_m(tz) + dz$. By Lemma 2.2(d), (3.1) and (3.2),

$$\wp(\ell_0) = (tz)^{2^m} + tz + d^2 z^2 + dz = v z^{2^m} + (t + d^2 + d)z + d^2 a = v(z^{2^m} + z) + d^2 a = v T_m(a) + d^2 a.$$

Put $\gamma = d^2a/v$, so that $\wp(\ell_0) = v(T_m(a) + \gamma)$. On the other hand $\delta(x) \in \langle 1, x \rangle$ by Lemma 3.5, so $\ell_0 = \delta(x)^{2^m} \in \{0, 1, u, u+1\}$ and $\wp(\ell_0) \in \{0, v\}$. Therefore $\varepsilon = T_m(a) + \gamma$ lies in $\mathbb{F}_2$, and $\varepsilon = 0$ exactly when $\delta(x) \in \{0, 1\}$. By Lemma 2.2(b) and (a),

$$\varepsilon = \text{Tr}(\varepsilon) = \sum_{i=0}^{m-1} \text{Tr}(a^{2^i}) + \text{Tr}(\gamma) = m \text{Tr}(a) + \text{Tr}(\gamma) = \text{Tr}(\gamma),$$

using $\text{Tr}(a) = 0$ from Lemma 3.4. By (3.2), $\gamma = tv/(d^2(d+v)) = v/d^2 + v/(d+v)$. By (3.1), $\rho(t/d^2) = v^2/(d+v)^2$, so $\text{Tr}(v/(d+v)) = \text{Tr}(t/d^2)$. Since $v+t = d^2+d$,

$$\text{Tr}(\gamma) = \text{Tr}\left(\frac{v+t}{d^2}\right) = \text{Tr}(1) + \text{Tr}\left(\frac{1}{d}\right) = 1 + \text{Tr}\left(\frac{1}{d}\right).$$

So $\delta(x) \in \{0, 1\}$ if and only if $\text{Tr}(1/d) = 1$. $\square$

4. PROOF OF THEOREM 1.1

The objects of Section 3 depend on x ; we write L_x , t_x , Φ_x , V_x and $d_x = x + x^{2^m}$ when the dependence matters. The second condition of Theorem 1.1 comes from the substitution $x \mapsto 1/x$.

Lemma 4.1. *For $x \in F \setminus \mathbb{F}_2$ and $y \in F$, $g(1/x, y/x) = x^{-(2^m+3)}g(x, y)$. Moreover $\Phi_{1/x}(y) = x^{-1}\Phi_x(xy)$, $V_{1/x} = x^{-1}V_x$ and $\delta(1/x) = x^{-1}\delta(x)$.*

Proof. With $u = x^{2^m}$,

$$u x^3 g(1/x, y/x) = u x^3 \left(\frac{y^{2^m}}{u} \cdot \frac{1+x}{x^2} + \frac{y^2}{x^2} \cdot \frac{x+u}{xu} + \frac{y}{x} \cdot \frac{u+x^2}{x^2u} \right) = g(x, y).$$

So $L_{1/x}(y) = x^{-(2^m+3)}L_x(xy)$. Since $t_{1/x} = x^{-2} + x^{-1} = t_x/x^3$,

$$\Phi_{1/x}(y) = \rho\left(\frac{L_x(xy)}{u t_x}\right) = \rho(u^{-1}) \Phi_x(xy) = x^{-1} \Phi_x(xy).$$

By the first statement of Lemma 3.5, applied to x and to $1/x$, $V_{1/x} = \{y \in F : L_{1/x}(\Phi_{1/x}(y)) = 0\} = \{y \in F : L_x(\Phi_x(xy)) = 0\} = x^{-1}V_x$. If $y_0 \in V_x \setminus \langle 1, x \rangle$, then $y_0/x \in V_{1/x} \setminus \langle 1, 1/x \rangle$, and $\delta(1/x) = \Phi_{1/x}(y_0/x) = x^{-1}\Phi_x(y_0) = x^{-1}\delta(x)$. $\square$

Proof of Theorem 1.1. By Lemma 3.5, $\dim Z(x) \in \{2, 3\}$, and $\dim Z(x) = 3$ if and only if $\delta(x) = 0$. Since $\delta(x) \in \langle 1, x \rangle$ and $x \neq 1$, $\delta(x) = 0$ if and only if $\delta(x) \in \{0, 1\}$ and $\delta(x) \in \{0, x\}$. By Lemma 3.6, the first condition is $\text{Tr}(1/d_x) = 1$. By Lemma 4.1, $\delta(x) \in \{0, x\}$ if and only if $\delta(1/x) \in \{0, 1\}$, which by Lemma 3.6 applied to $1/x$ means $\text{Tr}(1/d_{1/x}) = 1$. Finally $d_{1/x} = 1/x + 1/u = (x+u)/(xu)$. $\square$

5. COUNTING

Proof of Theorem 1.2. For $x \in F \setminus \mathbb{F}_2$ let $d = x + x^{2^m}$, $u = x^{2^m}$, $\alpha(x) = (-1)^{\text{Tr}(1/d)}$ and $\beta(x) = (-1)^{\text{Tr}(xu/d)}$. By Theorem 1.1,

$$N_n = \frac{1}{4} \sum_{x \in F \setminus \mathbb{F}_2} (1 - \alpha(x))(1 - \beta(x)) = \frac{1}{4} \left(q - 2 - \sum \alpha - \sum \beta + \sum \alpha\beta \right),$$

all sums over $x \in F \setminus \mathbb{F}_2$. By Lemma 2.2(c), $x \mapsto x + x^{2^m}$ maps $F \setminus \mathbb{F}_2$ two-to-one onto $T_0 \setminus \{0\}$. Hence

$$\sum \alpha = 2 \sum_{w \in T_0 \setminus \{0\}} (-1)^{\text{Tr}(1/w)} = \sum_{w \in F^*} (1 + (-1)^{\text{Tr}(w)}) (-1)^{\text{Tr}(1/w)} = -1 + K_n,$$

because $w \mapsto 1/w$ permutes F^* and $\sum_{w \in F^*} (-1)^{\text{Tr}(w)} = -1$. The map $x \mapsto 1/x$ permutes $F \setminus \mathbb{F}_2$, and $\beta(x) = \alpha(1/x)$ because $d_{1/x} = d/(xu)$; so $\sum \beta = K_n - 1$. Finally, $x \mapsto x + 1$ permutes $F \setminus \mathbb{F}_2$, leaves d unchanged and sends xu to $(x + 1)(u + 1) = xu + d + 1$; since $\text{Tr}(1) = 1$,

$$\beta(x + 1) = (-1)^{\text{Tr}((xu+1)/d)+1} = -\alpha(x)\beta(x).$$

So $\sum \alpha\beta = -\sum \beta = 1 - K_n$, and

$$N_n = \frac{1}{4}(q - 2 - 2(K_n - 1) + 1 - K_n) = \frac{q + 1 - 3K_n}{4}. \quad \square$$

Proof of Corollary 1.3. If $\dim Z(x) = 3$, then $A = Z(x)$ is a three-dimensional subspace containing 1 over which W_n sums to zero, by Lemma 2.3(a). Conversely, let $A \ni 1$ be such a subspace and $x' \in A \setminus \mathbb{F}_2$. Writing $A = \langle 1, x', y' \rangle$, Lemma 2.3(a) gives $g(x', y') = 0$, so $A \subseteq Z(x')$, and $Z(x') = A$ by Theorem 1.1. Hence the subspaces containing 1 with zero sum correspond to the elements x with $\dim Z(x) = 3$, six elements to each subspace, and there are $N_n/6$ of them.

Let $\mathcal{S}$ be the set of three-dimensional linear subspaces with zero sum. By Lemma 2.3(b), $A \in \mathcal{S}$ and $b \in F^*$ imply $bA \in \mathcal{S}$. The pairs (A, b) with $A \in \mathcal{S}$ and $b \in A \setminus \{0\}$ number $7|\mathcal{S}|$, and $(A, b) \mapsto (b^{-1}A, b)$ maps them bijectively to the pairs (B, b) with $B \in \mathcal{S}$, $1 \in B$ and $b \in F^*$, which number $(q - 1)N_n/6$. So $|\mathcal{S}| = N_n(q - 1)/42$. By Lemma 2.3(b), an affine subspace has zero sum exactly when its linear part does, and each linear part has 2^{n-3} cosets. The numbers are positive for $n \geq 7$ by the bound after Theorem 1.2, and $N_5 = 0$ because $K_5 = 11$. $\square$

Remark 5.1. For $n = 5$, Corollary 1.3 gives a second proof that $W_5 = X^7$ is third-order sum-free. For $n = 3$ the polynomial $g(x, Y)$ vanishes identically, and indeed $N_3 = 6 = 2^3 - 2$ since $K_3 = -5$.

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